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25 July 1979
SUPERSEDING
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6 March 1974

MILITARY SPECIFICATION

FILTERS, PARTICULATE: RECIRCULATION (550 CFM and 1200 CFM)

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of filters, particulate, recirculation for use in field protective shelters.

1.2 Classification. The particulate recirculation filters covered by this specification shall be of the following types as specified (see 6.2):

Type I - Entrance, Recirculation, Particulate Filter, 550 CFM
Type II - Shelter, Recirculation, Particulate Filter, 1200 CFM

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitations for bids or request for proposal form a part of this specification to the extent specified herein:

STANDARDS

FEDERAL

FED-STD-123 - Marking for Domestic Agencies (Civilian Agencies).

FSC 4240

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of US Military Property.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

C5-19-5209-20 - Filter, Particulate.
C5-19-5209-10 - Filter, Particulate.
DLA26-18-67 - Vibrating Machine, Filter Unit, Q110.
DLB76-2-639 - Penetrometer, Filter Testing, DOP, Q107.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on the applicable drawing and on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings:

Type I - C5-19-5209-20
Type II - C5-19-5209-10

3.1.2 Components. All components of this item shall conform to the specifications and drawings listed on the applicable drawings and their subsidiary drawings:

Type I - C5-19-5209-20
Type II - C5-19-5209-10

3.2 Manufacture and assembly. The particulate recirculation filters shall be assembled as specified below with subsidiary drawings:

Type I - C5-19-5209-20
Type II - C5-19-5209-10

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3.3 Airflow resistance.

3.3.1 Type I filter. The airflow resistance of the filter shall not exceed 0.750 inches water gage (w.g.) at a flow rate of 550 ± 30 cubic feet per minute (cfm) when tested as specified in 4.4.4.1.

3.3.2 Type II filter. The airflow resistance of the filter shall not exceed 0.750 inches w.g. at a flow rate of 1200 ± 60 cfm when tested as specified in 4.4.4.1.

3.4 Dioctylphthalate (DOP) smoke penetration.

3.4.1 Type I filter. The DOP smoke penetration shall not exceed 0.03 percent at the rated airflow when tested as specified in 4.4.4.1.

3.4.2 Type II filter. The DOP smoke penetration shall not exceed 15.0 percent at the rated airflow when tested as specified in 4.4.4.1.

3.5 Resistance to rough handling. After rough handling as specified in 4.4.4.2 for 15 minutes with the pleats, faces, and separators in the vertical position, the filters shall comply with the airflow resistance and DOP penetration requirements of 3.3 and 3.4.

3.6 Special marking. Filters shall be marked in accordance with MIL-STD-130 and include the DOP penetration, airflow resistance results, and serial number.

3.7 Preproduction. Prior to the start of regular production, a preproduction lot of particulate recirculation filters shall be produced in accordance with this specification for examination and test (see 4.3).

3.8 Workmanship. The filters shall be free from foreign matter (dirt, oil, or viscous material) and damage such as dented, bent, burned, or distorted frames; deformation or sagging of medium or separators; cracks in the adhesive; cracks or holes in the exposed portions of the medium; and damaged or improperly secured gaskets.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may utilize his own facility or any other

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facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Preproduction.

4.3.1 Sample. A preproduction lot of 10 particulate recirculation filters of each type shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production.

4.3.2 Inspection procedures.

4.3.2.1 For examination. Each particulate recirculation filter in the preproduction sample shall be examined for all requirements of the applicable drawings and this specification.

4.3.2.2 For tests. Each filter in the sample shall be tested in accordance with 4.4.4.

4.3.3 Acceptance/rejection criteria. Each filter in the preproduction sample shall comply with the requirements of this specification when examined and tested as specified in 4.3.2 to be acceptable. The supplier shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of each type of particulate recirculation filters produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions.

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However, no more than one lot of filter material shall be represented in any one lot of finished filters. Any change in equipment or procedure used in regular production which differs from that used in the preproduction sample shall be approved by the Government prior to its institution.

4.4.2 Sampling for examination and tests. Sampling for examination and tests and level A packaging shall be conducted in accordance with MIL-STD-105. For major defect 101, resistance to rough handling, level S-4 shall be used.

4.4.3 Inspection procedure.

4.4.3.1 For examination and tests. Sample particulate recirculation filters shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For critical defects. Each filter in the lot shall be tested for the critical defects listed in the classification of defects.

4.4.3.3 Classification of defects.

(a) Filters, particulate, (Dwg C5-19-5209-10,-20, as applicable).

Categories	Defects	Acceptance standards
Critical:		
1	Airflow Resistance	4.4.4.1
2	DOP Smoke Penetration	4.4.4.1
Major:	AQL 2.5 percent defective	
101	Resistance to rough handling	4.4.4.2
	AQL 1.0 percent defective	
102	Component missing	
103	Component incorrectly assembled	
104	Overall dimensions incorrect	
105	Marking incorrect	
106	Workmanship (3.8)	
Minor:	AQL 4.0 percent defective	
210	Marking missing or illegible	

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4.4.3.3 Classification of defects (continued).

(b) Preparation for delivery (section 5)

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Packaging box incorrect
102	Quantity per package incorrect
103	Packing box incorrect
104	Quantity of packages per pack incorrect
105	Marking missing or illegible

4.4.4 Tests.

4.4.4.1 Airflow resistance and DOP smoke penetration. The airflow resistance and DOP smoke penetration of the filters shall be determined with the Q107 DOP penetrometer (DLB76-2-639).

4.4.4.2 Resistance to rough handling. The filter shall be rough handled as specified for the required period of time with the Q110 Vibrating Machine (DLA26-18-67) and then tested for airflow resistance and DOP smoke penetration in accordance with 4.4.4.1.

5. PACKAGING

5.1 Packaging, level A. The particulate filters, type I and II shall be individually packaged as specified on the packaging data sheets for these items which are identified by their Federal stock number.

5.2 Packing, interplant shipment (see 6.5). The particulate filters, type I and II, packaged as specified in 5.1, shall be packed to prevent damage in shipment from the supply source to the receiving activity. Shipping containers shall comply with the rules and regulations applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, marking of packages (5.1), shall be specified on the packaging data sheet which is identified by the National stock number. Interplant shipment packs (5.2) shall be marked in accordance with FED-STD-123.

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5.4 Repair parts. When the filters, type I and II are procured for storage and issue as a repair part, preservation, packaging, packing, and marking shall be specified on the packaging data sheets which are identified by their National stock number.

6. NOTES

6.1 Intended use. The filters covered by this specification are used for the mechanical filtration of particulate matter from atmospheric air. These filters are components of the M51 shelter system.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Type filter required
(c) Level of packaging and packing required
(d) When packaging data sheet is required
(e) Preproduction

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

6.3 Manufacture. The following method of manufacture has been found to produce a satisfactory particulate filter. This procedure is furnished with the distinct understanding:

(a) That the contractor is not required to follow the method
(b) That the provisions of the specification must be met regardless of the method of manufacture
(c) That the Government makes no warranty that material made in accordance with this method will pass the specification requirements.

6.4 Manufacturing Procedure

(a) Filter material should be pleated on a suitable filter pleating machine using fiberboard separators to form and hold the pleats. A section of pleated material containing the number of folds which have been found to give the desired compactness should be tied-off and should be placed in a temporary U-frame. The exposed fiberboard separators should

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be removed and fluted aluminum separators inserted in their place. Any torn places in the filter material should be repaired with adhesive for filters. The pleated assembly shall be turned over and the operation repeated.

(b) The pleated section should be checked for proper compactness. Excessively tight packing will cause the fluted separators to cut into the filter material, and excessively loose packing will allow undesirable movement of the separators. Correct compactness may be obtained by removing or splicing in sections of pleated filter material to the ends of the assembly.

(c) Fiberboard backing strips should be inserted between each end of the pleated assembly and the temporary frame to keep pleats from bending over when trimmed. Both sides of pleated assembly should be trimmed with a high speed bench saw to obtain correct width. It is necessary to have an angle of no less than 30° between the saw blade and filter pleats when trimming to prevent the saw from tearing the paper.

(d) One coat of adhesive should be applied to both sides of the pleated assembly. Allow to penetrate and dry approximately two hours. A second coat of adhesive should be applied. This coat should be allowed to penetrate and dry approximately four hours. A third coat of adhesive should be applied and allowed to penetrate and dry approximately four hours. Additional coats should be applied if necessary to cover any pin holes which might arise when solvent evaporates. After careful inspection to see that there are no pin holes remaining, the excess adhesive should be trimmed from the edges of assembly and the temporary frame removed.

(e) The pleated assembly should be assembled in the frame and one coat of adhesive applied around the entire periphery sealing the frame to the filter. The adhesive should be allowed to dry two hours and a second coat applied and allowed to dry four hours. This operation should then be repeated on the reverse side.

(f) The final test for DOP penetration and air resistance should be performed and the filter marked.

6.5 Interplant shipment. Packing for interplant shipment is for supplies and materials that do not directly enter the military supply system. Typical interplant shipments are shipments from a subcontractor, vendor to a contractor, or a supplier to a military arsenal or a plant.

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6.6 International interest. Certain provisions of this specification are the subject of International Standardization Agreement QSTAG-451. When amendment, revision, or cancellation of this specification is proposed which affects or violates the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices if required.

Custodian:

Army - EA

Preparing activity:

Army - EA

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